

Chapter 7 — Swaps (Part 2)

J.C. Hull — *Options, Futures and Other Derivatives*

Structure of Part 2

This section focuses on:

1. Currency Swaps
2. Valuation of Interest Rate Swaps
3. Valuation of Currency Swaps
4. Swap Rates
5. Market Conventions
6. Swap Pricing Intuition

This part is mathematically and conceptually important because it connects:

- Swaps
 - Bonds
 - Forward contracts
 - Yield curves
 - Arbitrage pricing
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7.4 Currency Swaps

A currency swap is:

An agreement between two parties to exchange principal and interest payments in different currencies.

Unlike interest rate swaps:

Principal amounts are usually exchanged.

Main Purpose of Currency Swaps

Currency swaps are used to:

- Hedge exchange-rate risk
 - Obtain foreign currency financing
 - Access international capital markets
 - Reduce borrowing costs
 - Match foreign-currency cash flows
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Basic Structure of a Currency Swap

Typical structure:

Initial Exchange

Principal amounts exchanged at beginning.

Periodic Interest Payments

Each party pays interest in one currency.

Final Exchange

Principal amounts re-exchanged at maturity.

Example — Currency Swap

Suppose:

Company A

Needs:

- US dollars

Has comparative advantage in:

- British pound borrowing
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Company B

Needs:

- British pounds

Has comparative advantage in:

- Dollar borrowing
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Currency Swap Arrangement

Suppose:

- Exchange rate = 1 GBP = 1.60 USD
- Notional principals:
 - £10 million
 - \$16 million

At initiation:

- Company A gives £10 million
 - Company B gives \$16 million
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During Life of Swap

Company A:

- Pays dollar interest
- Receives pound interest

Company B:

- Pays pound interest
 - Receives dollar interest
-

At Maturity

Principals are re-exchanged:

- £10 million returned
 - \$16 million returned
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Key Difference Between Currency and Interest Rate Swaps

Feature	Interest Rate Swap	Currency Swap
Currency	Same	Different
Principal Exchange	Usually no	Usually yes
Main Risk	Interest rate risk	Exchange-rate + interest-rate risk

Why Principal Is Exchanged in Currency Swaps

Because each party actually requires:

- Funding in foreign currency

The swap effectively transforms:

Comparative Advantage in Currency Swaps

Currency swaps were historically motivated by:

Comparative borrowing advantages

Different firms may have:

- Better reputation in domestic markets
- Lower borrowing costs in home country
- Difficulty accessing foreign markets

Currency swaps help overcome these barriers.

Historical Importance of Currency Swaps

One famous early example:

IBM and the World Bank

The World Bank wanted:

- Swiss francs and Deutsche marks

IBM wanted:

- US dollar financing

A swap arrangement benefited both parties.

This transaction helped popularize currency swaps globally.

Risks in Currency Swaps

Currency swaps involve:

Exchange Rate Risk

Because principals are exchanged.

Interest Rate Risk

Because interest rates fluctuate.

Counterparty Credit Risk

Because swaps are OTC contracts.

7.5 Valuation of Interest Rate Swaps

One of the most important topics in swap markets is:

Swap Valuation

The value of a swap changes over time because:

- Interest rates change
 - Discount rates change
 - Future expected cash flows change
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Fundamental Pricing Principle

Hull shows that:

An interest rate swap can be viewed as:

- A long position in one bond
- A short position in another bond

This is one of the most important insights in derivatives pricing.

Fixed-for-Floating Interest Rate Swap

Suppose:

- Fixed-rate payer
- Floating-rate receiver

This position is equivalent to:

Long Floating-Rate Bond – Short Fixed-Rate Bond

Opposite Position

Floating-rate payer and fixed-rate receiver:

Long Fixed-Rate Bond – Short Floating-Rate Bond

Why This Works

Because:

- Fixed leg resembles fixed coupon bond

- Floating leg resembles floating-rate note

Thus swap valuation reduces to:

Bond valuation

Valuing a Fixed-Rate Bond

Value of fixed-rate bond:

$$B_{fixed} = \sum_{i=1}^n K e^{-r_i t_i} + L e^{-r_n t_n}$$

Where:

- K = coupon payment
 - L = principal
 - r_i = zero rate for maturity t_i
 - n = number of payments
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Valuing a Floating-Rate Bond

Immediately after payment date:

Floating-rate bond value $\approx$ par value

Reason:

- Coupon resets to market rate

Thus:

$$B_{floating} \approx L$$

where:

- L = notional principal
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Interest Rate Swap Value

For fixed-rate payer:

$$V_{swap} = B_{floating} - B_{fixed}$$

For floating-rate payer:

$$V_{swap} = B_{fixed} - B_{floating}$$

Example — Interest Rate Swap Valuation

Suppose:

- Notional principal = \$100 million
- Fixed rate = 5%
- Remaining life = 1.25 years
- Payments every 6 months

Assume current zero rates are known.

Step 1 — Calculate Fixed Bond Value

Discount all fixed payments:

$$B_{fixed} = \sum \text{PV of Coupons} + \text{PV of Principal}$$

Step 2 — Calculate Floating Bond Value

Floating bond is valued near:

100,000,000

plus accrued interest if necessary.

Step 3 — Compute Swap Value

For fixed payer:

$$V_{swap} = B_{floating} - B_{fixed}$$

Positive value:

- Swap is an asset

Negative value:

- Swap is a liability
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Important Insight

At initiation:

Swap value is approximately zero.

Why?

Because:

- Fixed swap rate is chosen so that:

$$B_{fixed} = B_{floating}$$

Swap Rates

The fixed rate that makes swap value zero at initiation is called:

Swap Rate

The swap rate depends on:

- Current zero curve
 - Payment frequency
 - Swap maturity
 - Market conditions
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Swap Rate Formula Intuition

The swap fixed rate is chosen so that:

$$\text{PV of Fixed Leg} = \text{PV of Floating Leg}$$

This is analogous to bond pricing.

Relationship Between Swaps and Bonds

Hull emphasizes:

Swaps are portfolios of bonds.

This insight simplifies:

- Pricing
 - Hedging
 - Risk management
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7.6 Valuation of Currency Swaps

Currency swaps are valued similarly.

However:

- Two currencies exist
 - Exchange rates matter
 - Both principal exchanges matter
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Currency Swap Valuation Principle

A currency swap can be viewed as:

Long Bond in One Currency – Short Bond in Another Currency

Example Structure

Suppose:

- Receive fixed USD interest
- Pay fixed EUR interest

Then value of swap:

$$V_{swap} = B_{USD} - S_0 B_{EUR}$$

Where:

- S_0 = current exchange rate
 - B_{USD} = value of dollar bond
 - B_{EUR} = value of euro bond
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Important Observation

Currency swap valuation requires:

- Interest rate curves in both currencies
- Current spot exchange rate
- Discounting cash flows separately

Currency Swap Exposure

Unlike interest rate swaps:

Currency swaps expose parties to exchange-rate fluctuations.

This creates larger market risk.

Market Conventions in Swaps

Swap markets follow standard conventions.

Payment Frequency

Typical frequencies:

Swap Type	Frequency
USD Swaps	Semiannual
EUR Swaps	Annual
Floating LIBOR	Quarterly/Semiannual

Day Count Conventions

Common conventions:

Leg	Convention
Fixed Leg	30/360
Floating Leg	Actual/360

Swap Confirmation

Swap agreements are documented using:

ISDA Agreements

ISDA:

International Swaps and Derivatives Association

These agreements standardize:

- Legal terms
 - Default provisions
 - Settlement procedures
 - Netting arrangements
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Netting in Swaps

Financial institutions often have many swaps with same counterparty.

Rather than exchanging all payments separately:

Netting is used.

Netting reduces:

- Credit exposure
 - Settlement costs
 - Operational risk
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Credit Risk in Swaps

Because swaps are OTC:

Counterparty default risk is important.

If one party defaults:

- Remaining positive value may be lost.
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Credit Exposure Changes Over Time

Swap exposure is dynamic.

Why?

Because swap value changes with:

- Interest rates
 - Exchange rates
 - Market conditions
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Mark-to-Market Value

The current value of the swap is called:

Mark-to-Market (MTM) Value

Positive MTM:

- Asset to one party
 - Liability to other party
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Hedging Swap Exposure

Swap dealers hedge exposure using:

- Bonds
 - Futures
 - FRAs
 - Interest rate options
 - Offset swaps
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Swap Curve

Modern markets use:

Swap Curve

instead of Treasury curve for many pricing applications.

The swap curve represents:

- Market swap rates across maturities

It is widely used as benchmark yield curve.

Why Swap Curve Became Important

Treasury yields may be distorted by:

- Government demand
- Liquidity effects
- Regulatory effects

Swap rates often better reflect:

- Interbank borrowing conditions
 - Corporate funding environment
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Important Conceptual Link

Swaps connect:

- Bond markets
- Money markets
- Forward markets
- Yield curves
- Credit markets

This is why swaps are central to modern derivatives theory.

Key Formulas from Part 2

Fixed Bond Value

$$B_{fixed} = \sum_{i=1}^n K e^{-r_i t_i} + L e^{-r_n t_n}$$

Floating Bond Approximation

$$B_{floating} \approx L$$

Interest Rate Swap Value

Fixed Payer

$$V_{swap} = B_{floating} - B_{fixed}$$

Floating Payer

$$V_{swap} = B_{fixed} - B_{floating}$$

Currency Swap Value

$$V_{swap} = B_{domestic} - S_0 B_{foreign}$$

Key Exam Concepts

You must understand:

- Currency swaps
 - Principal exchange
 - Swap valuation using bonds
 - Fixed vs floating bond interpretation
 - Swap rates
 - Swap curve
 - Netting
 - ISDA agreements
 - Mark-to-market valuation
 - Credit exposure
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Final Intuition of Part 2

Core ideas:

1. Currency swaps exchange cash flows in different currencies.
2. Interest rate swaps can be valued as combinations of bonds.
3. Fixed swap leg behaves like fixed-rate bond.
4. Floating swap leg behaves like floating-rate note.
5. Swap rates are chosen so initial value equals zero.
6. Swap valuation depends heavily on zero curves.
7. Counterparty credit risk is a central issue in OTC derivatives.
8. Swaps are deeply connected to modern fixed-income pricing theory.